

Chapter 10: Random Variability

Statistical Uncertainty, Super-Populations, and the Curse of Dimensionality

Applied Causal Inference Series — Use Case: "Did the Retention Offer Actually Reduce Churn?"

Executive Summary

Chapters 1–9 examined causal inference under the assumption of an effectively infinite sample size ("Olympian certainty"), isolating systematic biases: **confounding**, **selection bias**, and **measurement error**. Chapter 10 introduces **random variability**—the inescapable statistical uncertainty arising from analyzing finite samples of real-world data.

In CRM and retention analysis, even when an A/B test or observational study is perfectly designed with zero systematic bias, finite sample size (n) means that observed differences in churn rates can still occur by chance. This chapter establishes the mathematical bridge between **identification** (can we compute the causal effect with infinite data?) and **estimation** (how do we quantify uncertainty with finite data?), introducing confidence intervals, the super-population framework, the conditionality principle, and the curse of dimensionality.

1 10.1 Identification vs. Estimation

(Section 10.1, pp. 131–134)

In causal inference, it is vital to distinguish between two fundamental tasks:

- **Identification Problem (Chapters 1–9):** Asking whether the parameter of interest—the *estimand* $RD = \Pr[Y^{a=1} = 1] - \Pr[Y^{a=0} = 1]$ —can be uniquely expressed in terms of the observed data distribution $P(Y, A, L)$ under untestable causal assumptions (exchangeability, positivity, consistency). Identification assumes $n \rightarrow \infty$.
- **Estimation Problem (Chapter 10 onward):** Constructing a function of finite sample data—an *estimator* $\widehat{RD}$ —to calculate a numerical *estimate* and quantify its statistical uncertainty due to sampling variability.

CRM Use Case: The Small Sample Dilemma

Suppose a pilot retention campaign targets $n = 20$ high-risk customers: 13 receive a retention discount ($A = 1$) and 7 receive no offer ($A = 0$). Suppose 7 out of 13 treated customers churn ($Y = 1$), while 3 out of 7 untreated customers churn.

$$\widehat{\Pr}[Y = 1 \mid A = 1] = \frac{7}{13} \approx 0.538, \quad \widehat{\Pr}[Y = 1 \mid A = 0] = \frac{3}{7} \approx 0.429 \implies \widehat{RD} = 0.538 - 0.429 = +0.109$$

Does this mean the offer *increased* churn by 10.9 percentage points? With $n = 20$, random sampling fluctuations can easily produce an observed risk difference of +10.9% even if the true population effect is zero or negative. Estimation provides confidence intervals to quantify this range of uncertainty.

Consistency and Unbiasedness of Estimators

An estimator $\hat{\theta}_n$ is **consistent** if it converges in probability to the true parameter θ as sample size $n \rightarrow \infty$. To construct valid 95% Wald confidence intervals ($\hat{\theta} \pm 1.96 \cdot \widehat{\text{SE}}$), statisticians require the estimator to be **Uniformly Asymptotic Normal and Unbiased (UANU)** (*Technical Point 10.1*).

2 10.2 Quantifying Statistical Uncertainty

(Section 10.2, pp. 134–136)

Statistical theory provides tools to express precision around point estimates:

- **Standard Error (SE):** Quantifies the expected dispersion of the point estimate across repeated random samples of size n . For a sample proportion $\hat{p} = \widehat{\Pr}[Y = 1 \mid A = 1]$, $\widehat{\text{SE}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.
- **95% Wald Confidence Interval:** $\hat{\theta} \pm 1.96 \cdot \widehat{\text{SE}}(\hat{\theta})$. In large samples, 95% of such intervals calculated across repeated studies will contain the true super-population parameter.
- **Frequentist Interpretation:** A 95% confidence interval from a single sample does *not* mean there is a 95% probability that the true parameter lies inside that specific interval; the fixed parameter is either inside or outside. The 95% refers to the long-run coverage frequency.

Fine Point 10.2: "Compatibility Intervals" and Systematic Bias

Standard confidence intervals **only reflect random error**. They assume zero unmeasured confounding, zero selection bias, and zero measurement error. As sample size $n \rightarrow \infty$, the standard error shrinks to zero, making the confidence interval infinitely narrow—yet systematic bias remains unchanged!

For this reason, researchers increasingly refer to confidence intervals as **compatibility intervals** (Amrhein et al., 2019; Greenland, 2019), emphasizing that they only show which parameter values are statistically compatible with the data *given* all structural adjustment assumptions.

3 10.3 The Myth of the Super-Population

(Section 10.3, pp. 136–138)

Reporting standard binomial confidence intervals requires a theoretical foundation for where randomness originates. *What If* outlines two scenarios:

1. **Scenario 1 (Super-Population Framework):** The n study subjects are a random sample drawn from a near-infinite target **super-population** (e.g., all current and future subscribers of the company). The estimand is the super-population probability $\Pr[Y^a = 1]$.
2. **Scenario 2 (Stochastic Counterfactuals in Sample):** The n study subjects are the entire population of interest, but each individual i has an inherent stochastic counterfactual probability p_i^a (like a quantum coin flip).

Because individual risk p_i^a varies across customers due to unobserved baseline traits (genetics, income, habits), Scenario 2 leads to conservative, non-calibrated intervals for sample averages. Therefore, any analysis reporting standard confidence intervals implicitly adopts **Scenario 1**: treating the dataset as a sample from a hypothetical super-population of potential customers.

4 10.4 The Conditionality Principle & Chance Imbalances

(Section 10.4, pp. 138–142)

Consider an A/B test with $n = 240$ subjects randomly assigned to receive a retention offer ($A = 1, n_1 = 120$) or control ($A = 0, n_0 = 120$). The unadjusted risk difference is $\widehat{RD}_{\text{unadj}} = -0.15$ (95% CI: $[-0.26, -0.04]$).

Later, analysts discover that baseline account activity L (1 = low activity, 0 = high activity) was accidentally imbalanced by chance: 80 low-activity users ended up in $A = 1$, but only 40 in $A = 0$.

Stratum	Treated ($A = 1$)	Control ($A = 0$)	Risk Diff (RD_L)	Adjusted Estimate
High Activity ($L = 0$)	4 / 40 (10%)	2 / 80 (2.5%)	+7.5%	Standardized RD: 0.000 (95% CI: $[-0.076, +0.076]$)
Low Activity ($L = 1$)	20 / 80 (25%)	40 / 40 (100%)	−75.0%	

Table 1: Chance imbalance of baseline covariate L in a randomized trial.

The Conditionality Principle

The **Conditionality Principle** states that inference about a causal effect should be performed **conditional on ancillary statistics**—statistics like the observed baseline sample distribution $\widehat{\Pr}[A = 1 \mid L]$ that contain no information about the causal parameter itself but quantify chance sample imbalances.

Even in a randomized experiment where $A \perp\!\!\!\perp L$ in the super-population, **the adjusted (ML/standardized) estimator is conditionally unbiased**, whereas the unadjusted estimator is conditionally biased by chance sample imbalances.

5 10.5 The Curse of Dimensionality

(Section 10.5, pp. 142–144)

When adjusting for a high-dimensional vector of baseline covariates $L = (L_1, L_2, \dots, L_{100})$, the number of joint strata is $2^{100} \approx 1.27 \times 10^{30}$ —far exceeding any real-world sample size n .

- **Nonparametric Failure:** Most strata contain 0 or 1 subject. Nonparametric conditional estimation breaks down; conditional risk differences cannot be computed in empty cells, yielding completely uninformative confidence intervals $[-1, +1]$.
- **The Need for Models (Part II Motivation):** To overcome the curse of dimensionality, analysts must substitute non-parametric stratification with **parametric causal models** (logistic regression, IP weighting models, g-computation, g-estimation), borrowing information across strata by making functional form assumptions.

Summary Comparison: Systematic Bias vs. Random Variability

Property	Systematic Bias (Ch. 7–9)	Random Variability (Ch. 10)
Source	Structural (Confounding, Selection, Measurement)	Sampling fluctuations due to finite sample size
Behavior as $n \rightarrow \infty$	Persists completely (does not shrink)	Shrinks to zero (Law of Large Numbers)
Mitigation	Causal design, DAGs, G-methods adjustment	Larger sample sizes, confidence intervals, SE
Quantified By	Sensitivity / Quantitative Bias Analysis	95% Wald / Bootstrap Confidence Intervals